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School of Computer Science and Artificial Intelligence

Lab Assignment # 16.2

Program	**: B. Tech (CSE)**
Specialization	**: -**
Course Title	**: AI Assisted Coding**
Course Code	**: 23CS002PC304**
Semester	**II**
Academic Session	**: 2025-2026**
Name of Student	**: M.Srinath**
Enrollment No.	**: 2403A51L29**
Batch No	**: 51**
Date	**: 17/03/26**

Task 1: (Design Database Schema for Student Management System)

Input:

Schema SQL

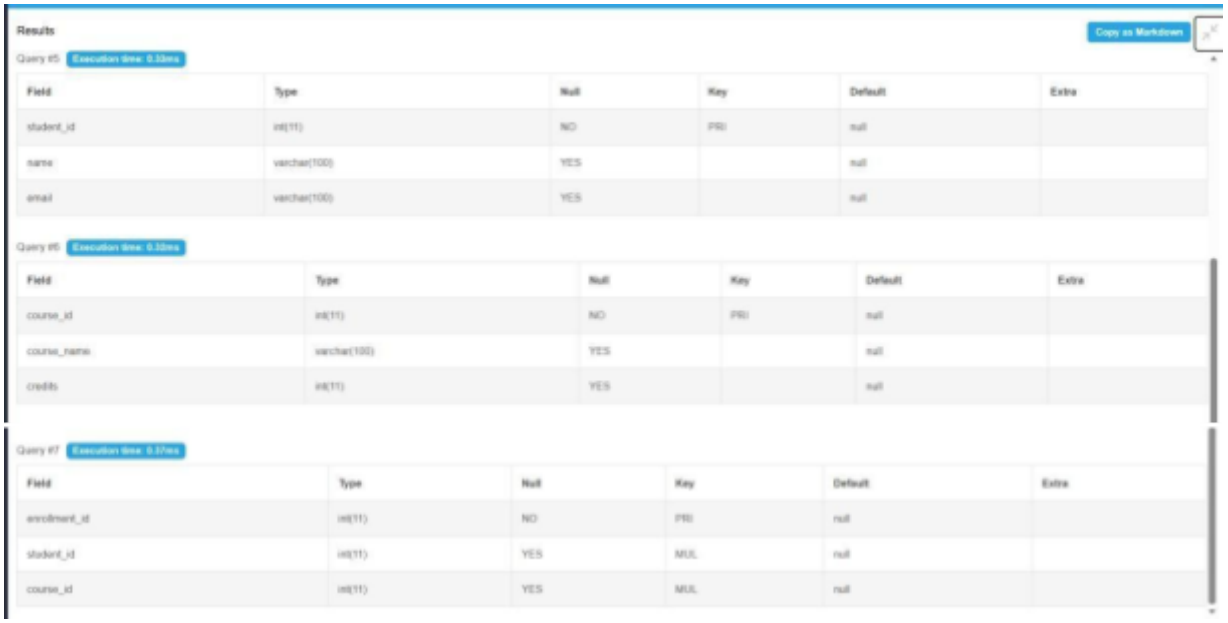
```sql
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(100),
    email VARCHAR(100)
);

CREATE TABLE Courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(100),
    credits INT
);

CREATE TABLE Enrollments (
    enrollment_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);

-- SHOW OUTPUT
SHOW TABLES;
DESCRIBE Students;
DESCRIBE Courses;
DESCRIBE Enrollments;
```

Output:



Explanation:

1. Defines structure of database using tables: Students, Courses, Enrollments.
2. Primary keys ensure unique identification of each record.
3. Foreign keys maintain referential integrity between related tables.
4. Follows normalization (1NF, 2NF, 3NF) to remove redundancy and ensure consistency.

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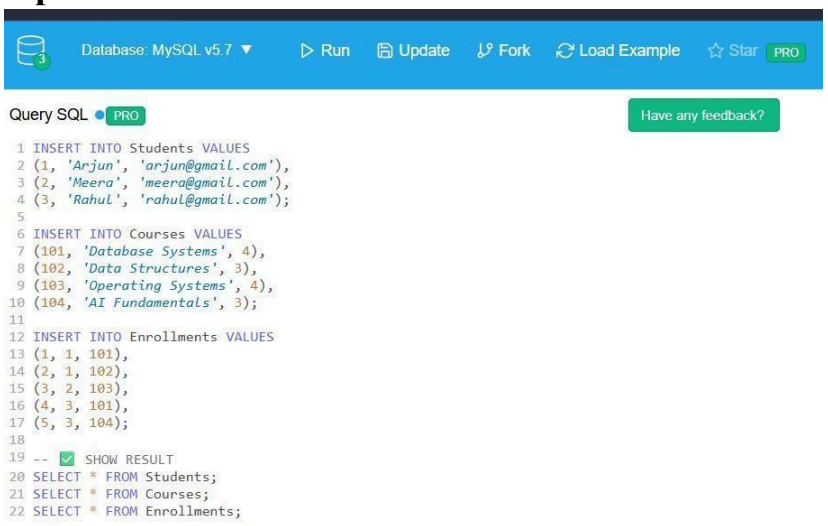
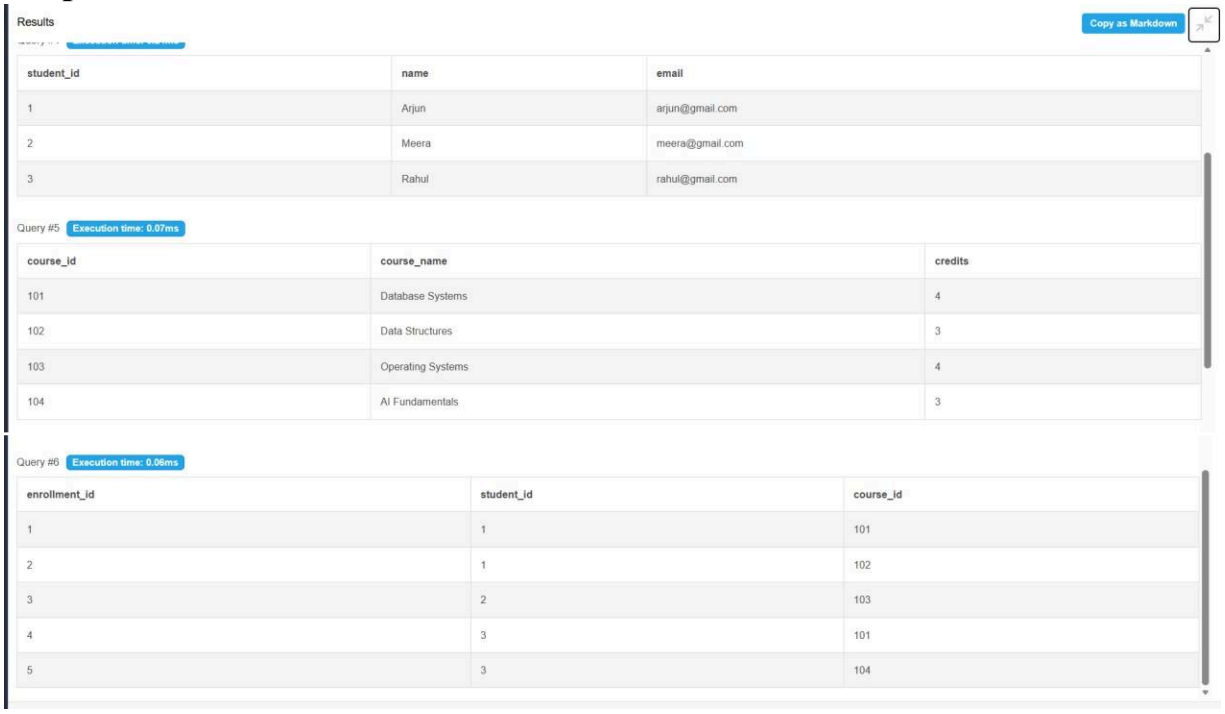
Task 2: (Insert Sample Records)

Input:



Output:



Explanation:

1. Populates tables with initial data using INSERT statements.
2. Ensures relational links by matching student_id and course_id.
3. Helps in testing database functionality with sample dataset.
4. SELECT queries verify whether data insertion is successful.

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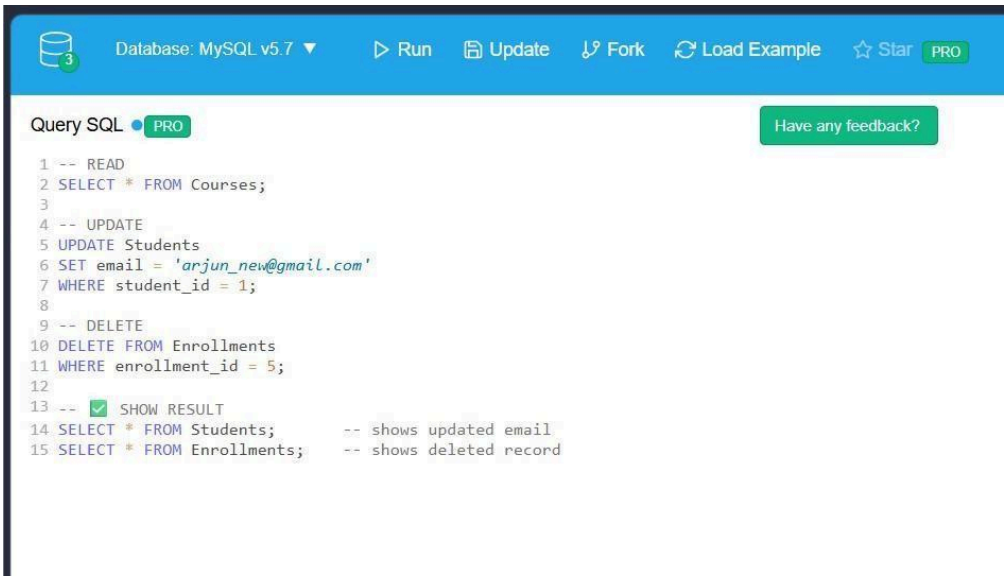
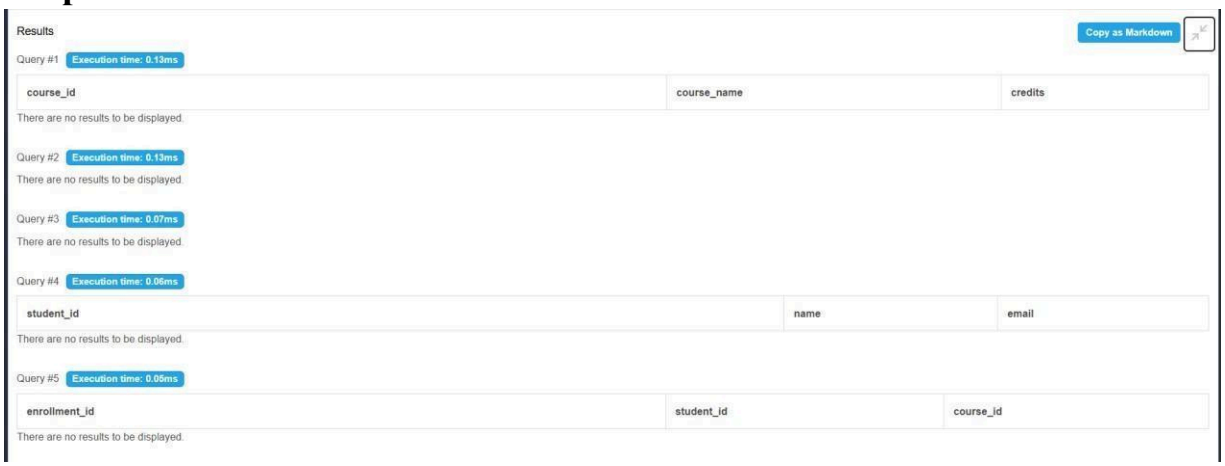
Task 3: CRUD Operations

Input:



Output:



Explanation:

1. SELECT retrieves data from tables for viewing records.
2. UPDATE modifies existing data like student email.
3. DELETE removes specific records such as enrollments.
4. Validates operations using SELECT to ensure correctness.

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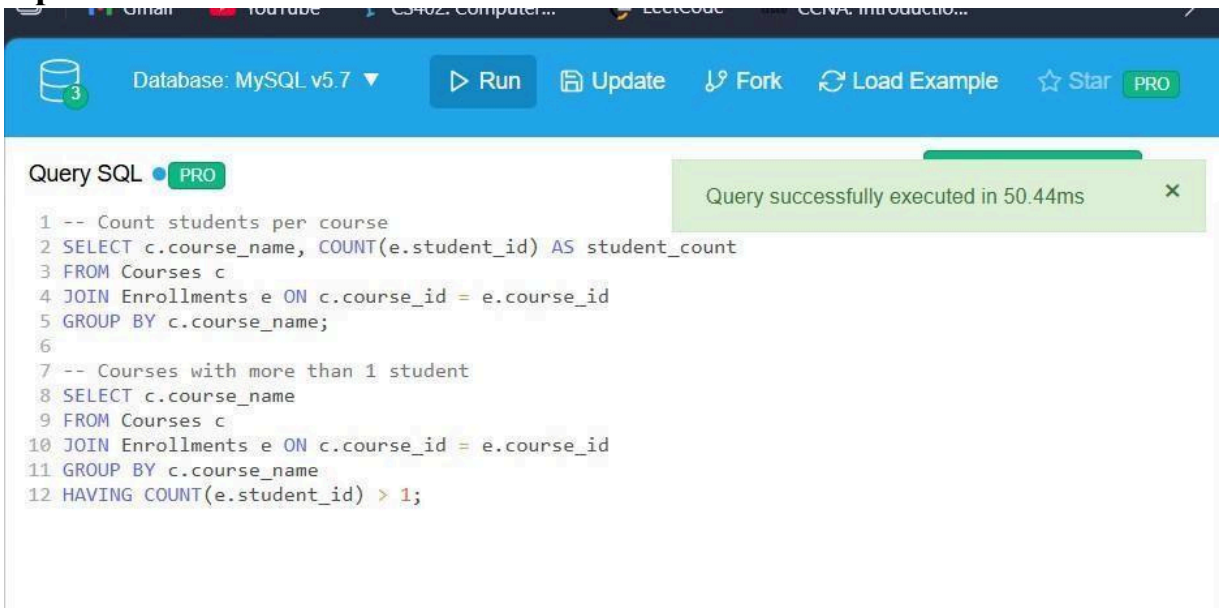
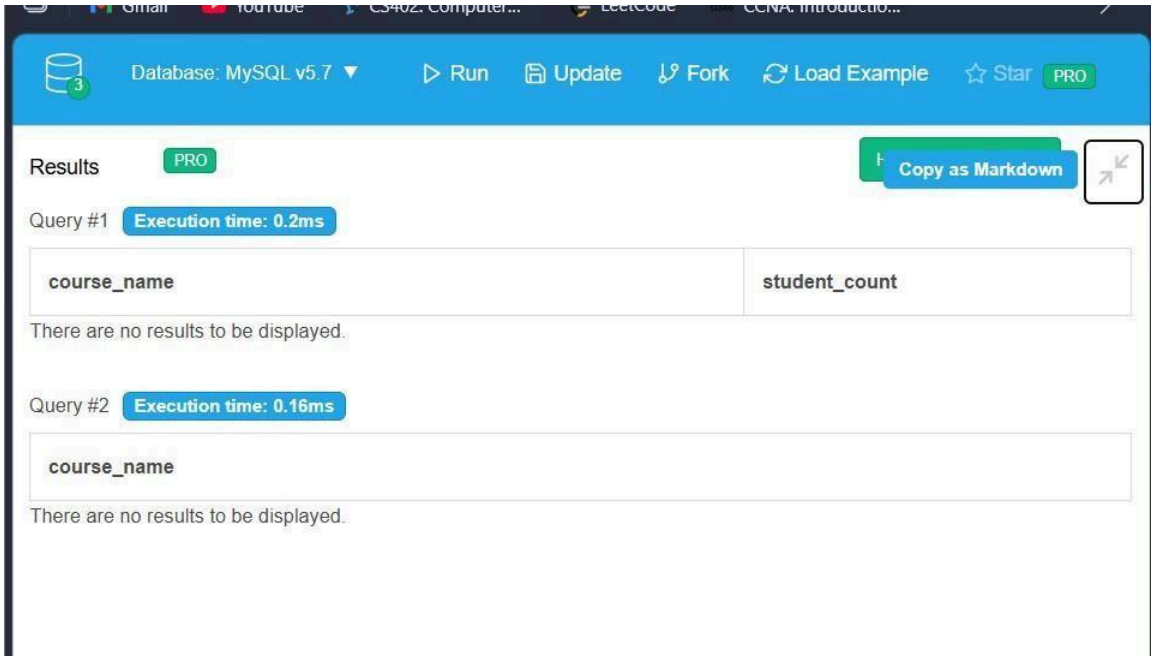
Task 4: Aggregate Queries

Input:



Output:

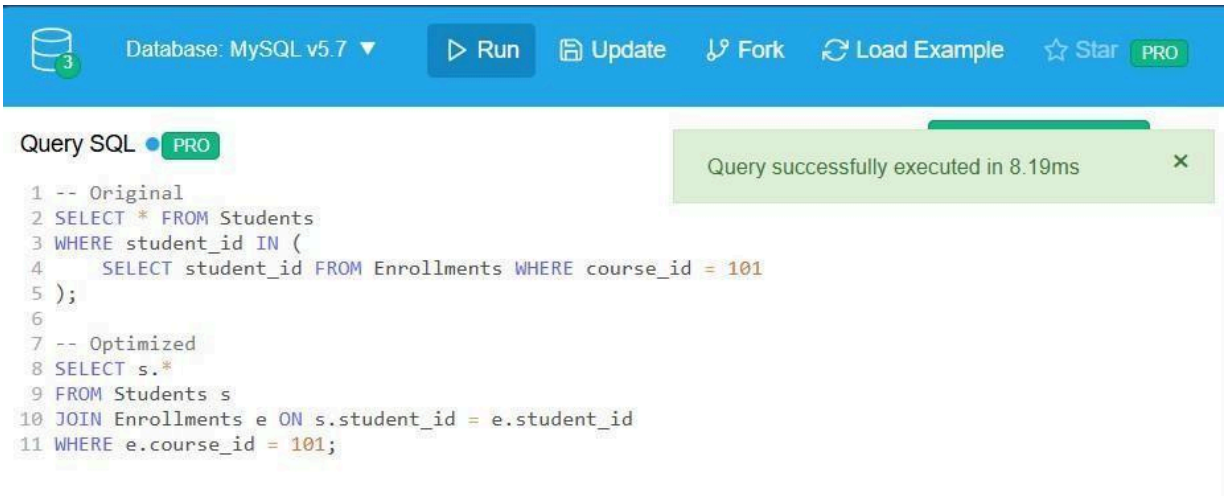


Explanation:

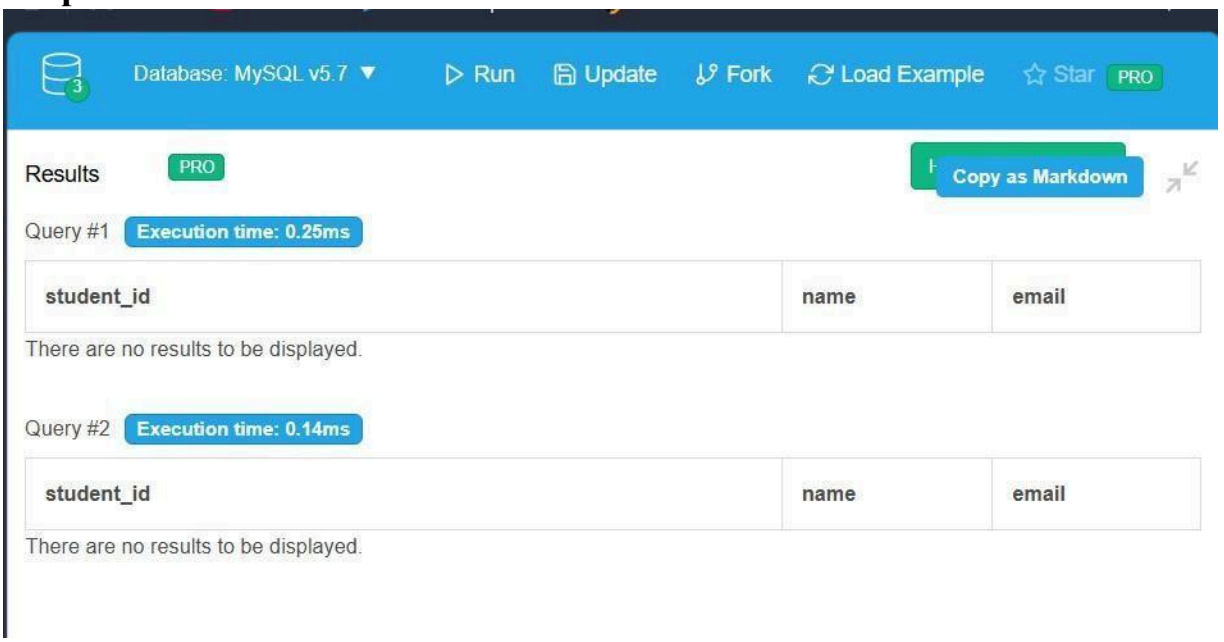
1. Uses functions like COUNT() to summarize data.
2. GROUP BY groups records based on course names.
3. HAVING filters grouped data (e.g., courses with >1 student).
4. Helps in analysis like enrollment trends per course.

Task 5: Query Optimization

Input:



Output:



Explanation:

1. Identifies inefficient queries (e.g., subqueries).
2. Rewrites queries using JOINs for better performance.
3. Ensures both original and optimized queries give same result.
4. Improves execution speed, especially for large datasets.